EXPLORATORY DATA ANALYSIS

WEEK-3

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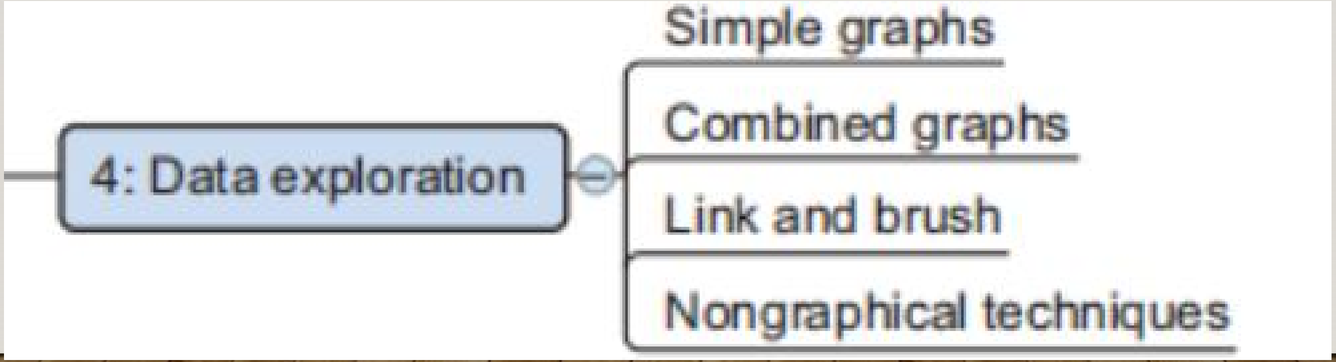
EDA-UNDERSTANDING YOUR DATA BEFORE BUILDING MODELS

- Information becomes much easier to grasp when shown in a picture, graphical techniques are used to gain an understanding of your data and the interactions between variables.
- The goal isn't to cleanse the data, but it's common that you'll still discover anomalies you missed before, forcing



WHY IS EDA IMPORTANT?

- Understand data distribution and structure
  - Who constructed this dataset, when and why?
  - How big is it ?
  - What do the fields mean ?
- Identify data cleaning needs
  - Guide feature engineering
  - Choose appropriate models and methods
  - Save time by avoiding poor model choices

EDA WORKFLOW

1. Data Collection & Import
2. Data Inspection
3. Data Cleaning
4. Univariate Analysis (Single Variable)
5. Bivariate/Multivariate Analysis
6. Outlier Detection
7. Feature Engineering (Preliminary)
8. Visualization

UNIVARIATE ANALYSIS

- Focus: One variable at a time
- If we have numerical variable, we analyse it through Distribution, summary stats (mean, median, std, quartiles).
- If we have categorical variable, we analyse it by Frequency counts and bar plots.
- Techniques: Summary stats, Histograms, Boxplots, Value counts

BIVARIATE ANALYSIS

- Focus: Relationship between two or more variables
- Correlations between numerical variables.
- Cross-tabulations for categorical variables.
- Techniques: Scatter plots, Bar plots, Correlation matrix, Cross-tabulations and heatmaps to check relationships, Pair plots, Grouped bar charts

OUTLIERS

- Detection
  - Statistical rules (IQR, Z-score).
- Visualization
  - Boxplots, histograms
- Deal with Outliers
  - Transform and Impute, cap, or remove.

Z-SCORE

Z-score tells us how many standard deviations a value is away from the mean.

If |Z| > 3, the value is usually considered an outlier.

$$Z = \frac{x - \mu}{\sigma}$$

IQR (INTERQUARTILE RANGE) METHOD

- IQR measures the **spread of the middle 50% of data**.
- Outliers are values outside the range:
- **Steps**
  - Find **Q1** (25th percentile) and **Q3** (75th percentile).
  - Compute **IQR = Q3 - Q1**.
  - Define lower & upper bounds.
  - Any value outside bounds = Outlier.

$$[Q1 - 1.5 \times IQR,\ Q3 + 1.5 \times IQR]$$

FEATURE ENGINEERING

- Create new variables (ratios, groupings, transformations).
- Encode categorical variables (label encoding, one-hot encoding).

You can apply Feature Scaling, Min-Max, and Z-Score to normalize data and apply each technique in Python using pandas' methods.

Several approaches for normalization:

① $x_{new} = \frac{x_{old}}{x_{max}}$

Simple Feature scaling

② $x_{new} = \frac{x_{old} - x_{min}}{x_{max} - x_{min}}$

Min-Max

③ $x_{new} = \frac{x_{old} - \mu}{\sigma}$

Z-score

Z-SCORE SCALING

Standard Scaling (Z-score Normalization)

- Transforms data to have mean = 0 and standard deviation = 1
- Formula:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

A **z-score** is a statistical measure that tells you **how many standard deviations a data point is away from the mean** of a dataset.

Code Example:

```
scaler = StandardScaler()
scaled_standard = scaler.fit_transform(df)

df_standard = pd.DataFrame(scaled_standard, columns=df.columns)
print("\nStandard Scaled Data:")
print(df_standard)
```

FEATURE SCALING

Data Normalization

age	income
20	100000
30	20000
40	500000

age	income
0.2	0.2
0.3	0.04
0.4	1

Not-normalized

- "age" and "income" are in different range.
- hard to compare
- "income" will influence the result more

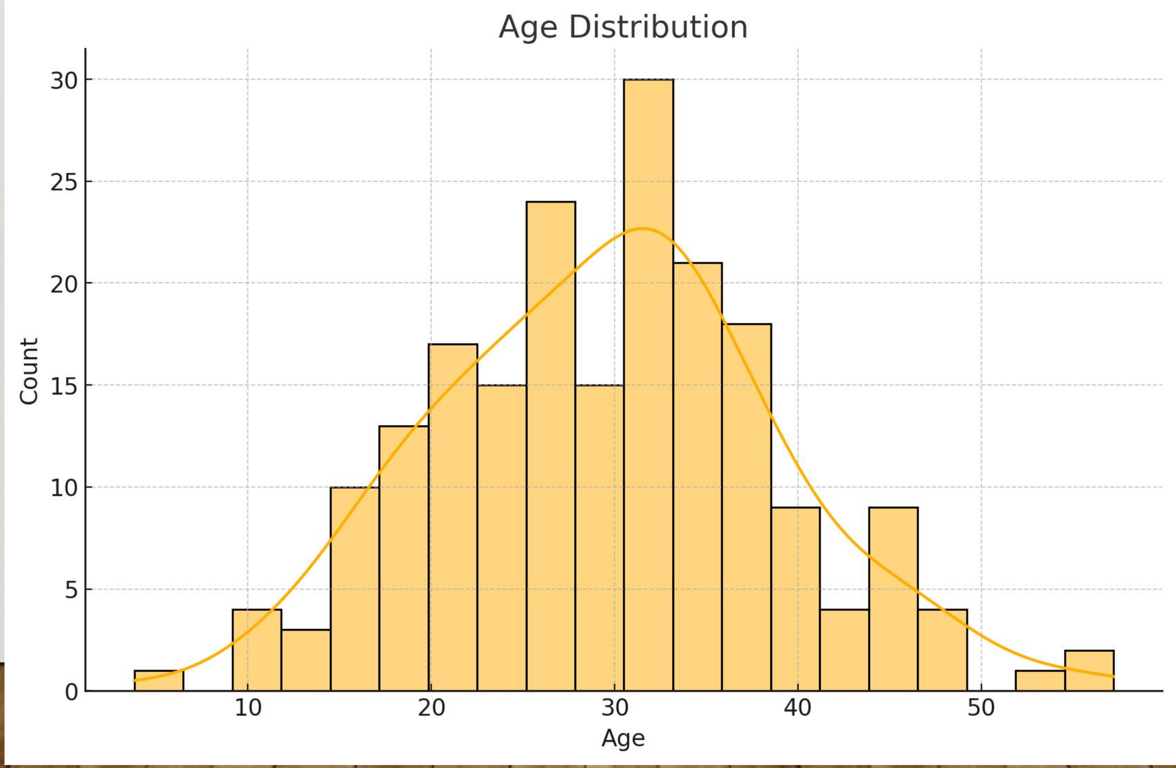
Normalized

- similar value range.

Data normalization helps make variables comparable and helps eliminate inherent biases in statistical models.

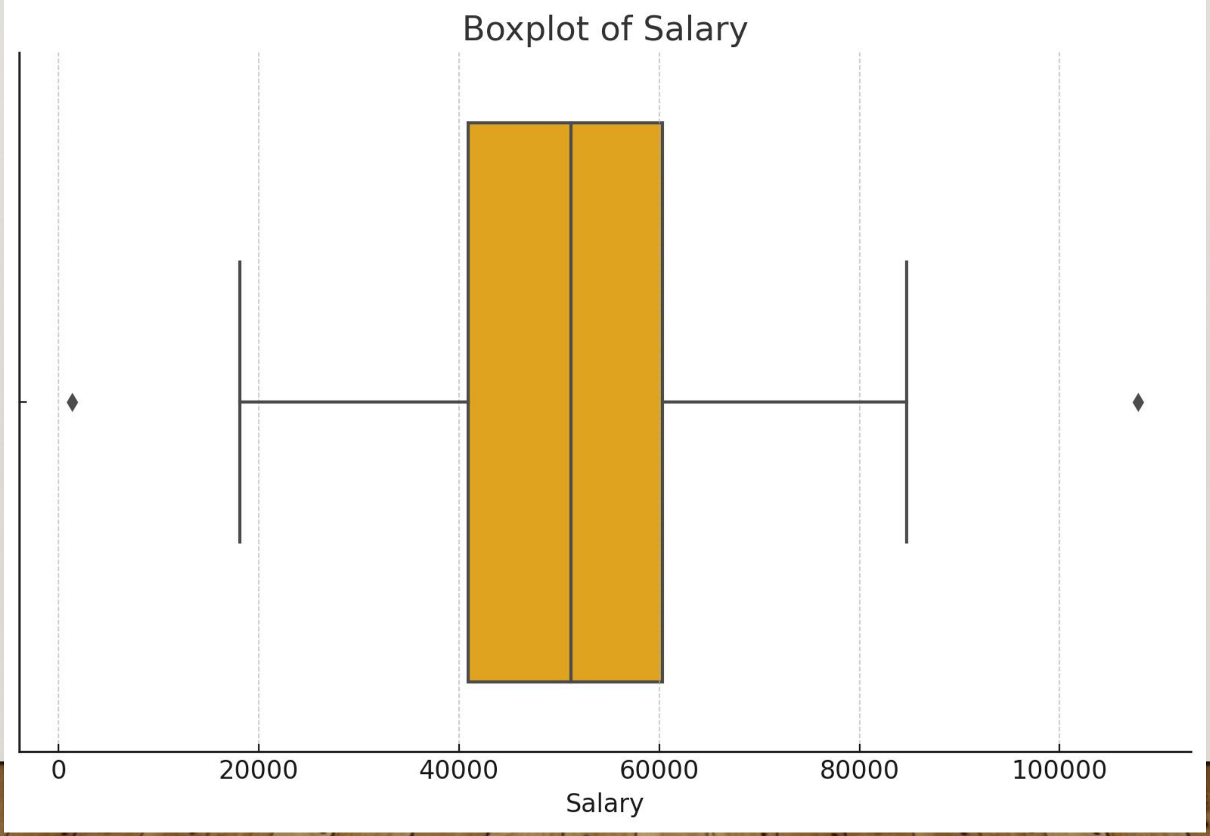
HISTOGRAM: AGE DISTRIBUTION

In a histogram a variable is cut into discrete categories and the number of occurrences in each category are summed up and shown in the graph.

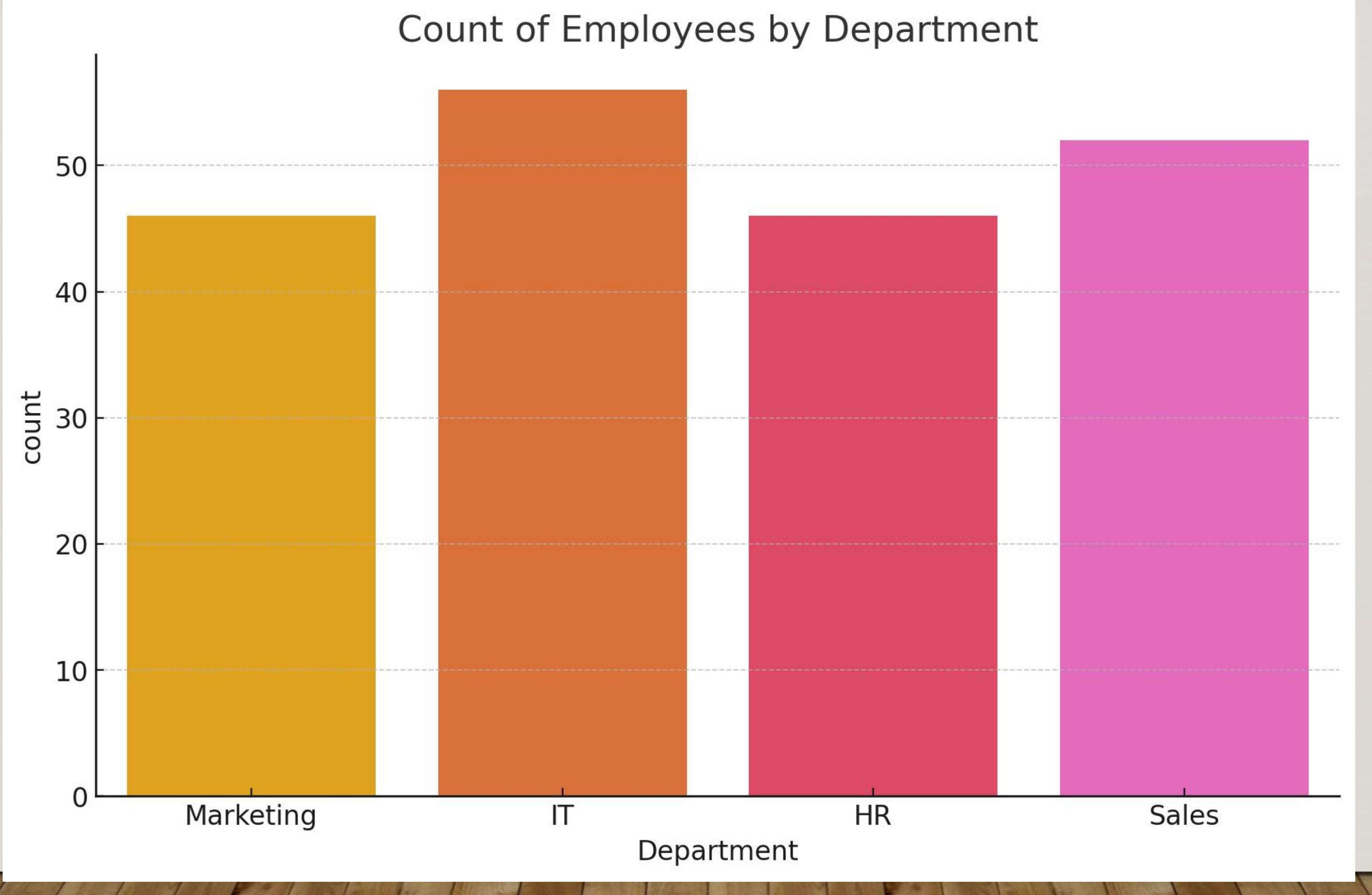


BOXPLOT: SALARY OUTLIER DETECTION

The boxplot doesn’t show how many observations are present but does offer an impression of the distribution within categories. It can show the maximum, minimum, median, and other characterizing measures at the same time.

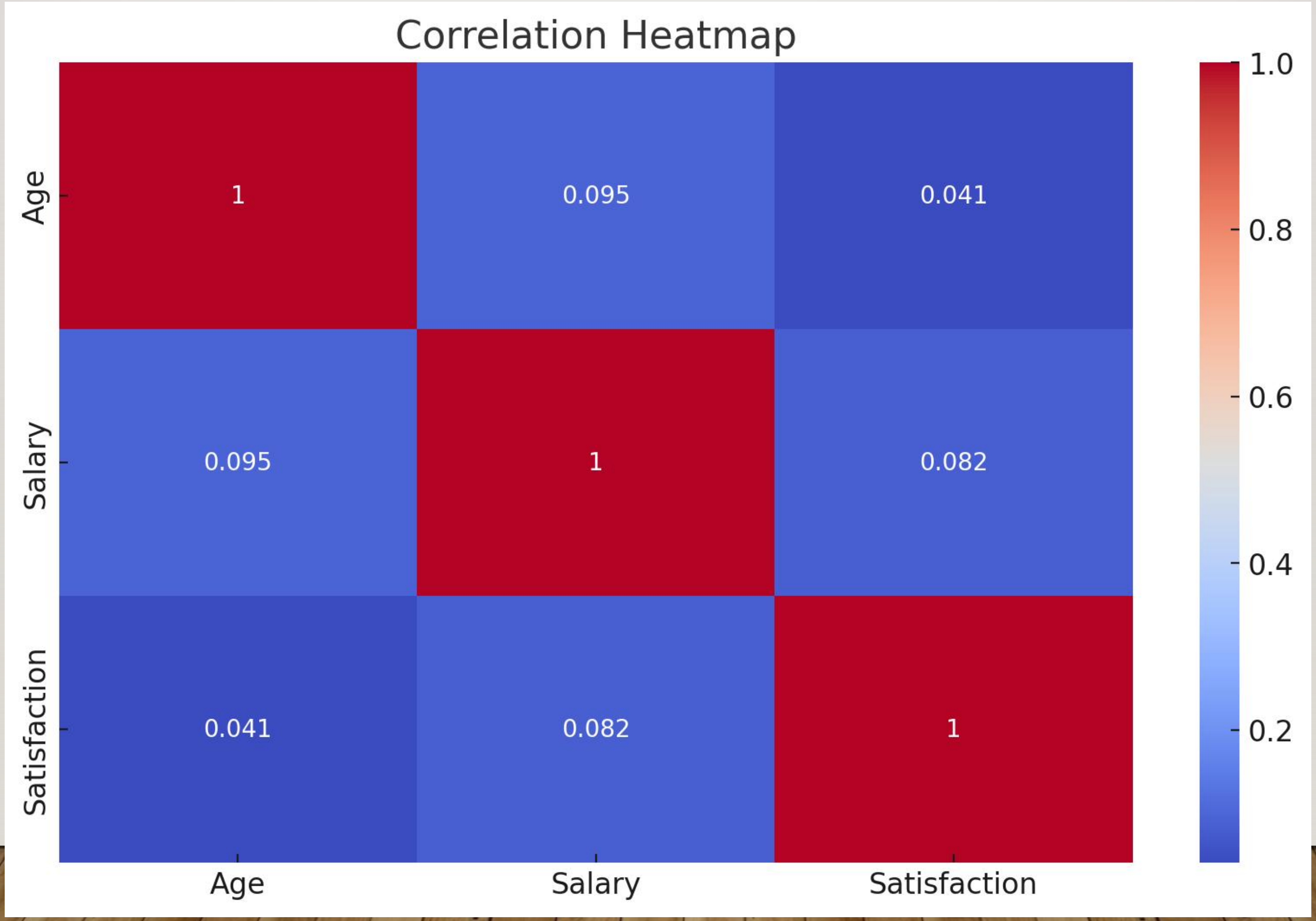


BAR PLOT: EMPLOYEE COUNT BY DEPARTMENT



SCATTER PLOT: AGE VS SALARY BY GENDER

HEATMAP: CORRELATION MATRIX



BRUSHING AND LINKING

- With brushing and linking you combine and link different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs.
- Link and brush allows you to select observations in one plot and highlight the same observations in the other plots.

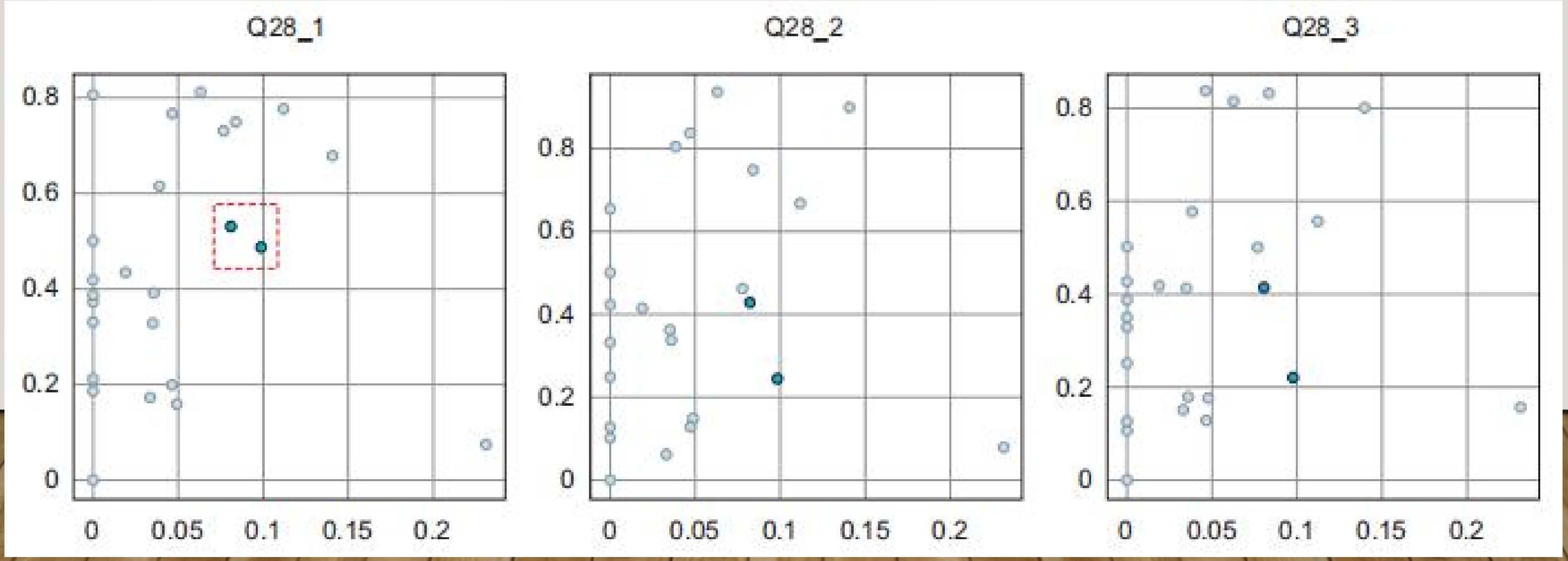


CHART SUGGESTIONS

Comparison-**over time across categories/items**

Composition-how parts make up a whole

What would you like to show???

Distribution-Explore how values are spread

Relationship-How variable relate or corelate?

COMPARISON

- Do you want to compare data **over time** or **across categories/items**?
- **Over Time**: Are there few or many time periods?
  - *Few*: Use simple line or column charts.
  - *Many*: Consider area charts or sparklines.
- **Across Items**:
  - Single variable: Bar chart.
  - Multiple variables: Side-by-side or grouped bar charts.

COMPOSITION

- Are you showing how parts make up a whole?
- **Static composition**: Pie / Donut charts.
- **Over time**: Stacked area or stacked bar charts to illustrate how parts accumulate or change.

DISTRIBUTION

- Do you want to explore how values are spread?
- For **single numeric variable**: Histogram or bar chart.
- For **two variables**: Box plots, scatter plots.

RELATIONSHIP

- Are you showing how variables relate or correlate?
- **Two variables**: Scatter plot.
- **Three variables**: Bubble chart or coloured scatter with a third dimension.